

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Code of Conduct:

I J. Kirthiga / 192421265 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

J. Kirthiga
Signature of the student:

20
20
20/11

QUESTIONS:4

Total Marks:40

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS, DNS pre-fetching, and TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CO-3 ASSESSMENT TOOL-2

CASE STUDY

Root Cause Analysis: The issue is mainly due to UDP transport protocol along with weak application-level buffering

- video conferencing uses UDP for low latency
- UDP does not retransmit lost packets

Given:

8% packet → missing audio/video data

180 ms jitter → irregular packet arrival

This causes voice distortion and frozen frames

At the same time:

Poor jitter buffering at the application layer

increases delay and instability

The problem is primarily due to UDP limitations, worsened by inefficient application banding.

Justification (Real-time Requirements)

Real time app needs:

- low latency → UDP is preferred over TCP
- No retransmission delays
- smooth playback

TCP is not suitable because

- Retransmission causes delay

UDP is suitable but:

- cannot handle packet loss and jitter alone

3) Recommended Improvements

Transport level

- Use RTP over UDP (adds sequence numbers, timestamps)
- Use RTCP for monitoring network conditions

Application - level:

- Adaptive-jitter buffer $\rightarrow$ smooth playback
- forward Error correction $\rightarrow$ FEC $\rightarrow$ recover lost data
- Adaptive bitrate $\rightarrow$ adjust quality based on bandwidth
- Packet loss concealment $\rightarrow$ reduce distortion

Issue	Impact
Packet loss(8%)	Missing frames
High jitter	uneven playback
No reliability	Distortion

So UDP is worst choice, but needs support from application layer

Conclusion:

- problem is mainly due to UDP packet loss and jitter
- Solution requires :
 - RTP / RTCP (transport)
 - buffering & adaptive techniques

Reasons for High DNS Resolution time

The high DNS lookup time (800ms) is mainly due to

- Geographic distance between users and DNS Servers
- Lack of coaching on Improper TTL Settings, causing repeated queries
- Multiple Recursive lookups (root $\rightarrow$ TLD $\rightarrow$ authoritative Servers)
- Server overload due to centralized DNS Infrastructure
- No Anycast routing, so users are not directed to the nearest Server

Proposed DNS Architecture (<50 ms)

To reduce lookup time:

- Anycast DNS

• Deploy DNS Servers globally with the same IP address
So users connect to the nearest server, reducing latency

- DNS pre-fetching: Resolve domain names in advance to avoid delay during actual access

- TTL optimization - Use Higher TTL for stable services and moderate TTL for dynamic services to improve caching efficiency

Additional Improvements: Use local DNS resolvers and load balancing to further reduce delays

Advantages:

- High performance due to reduced latency and caching
- Scalability as more DNS Servers can be added globally
- High Availability since traffic is redirected if a server fails.

Limitations:

- complex setup for Anycast Implementation
- Higher cost due to Global Infrastructure
- TTL trade-off between caching efficiency and update speed
- Routing dependency on ISP policies

Anycast DNS (core solution)

- Deploy multiple DNS Servers Globally
- All Servers share the same IP address
- Routing directs users to nearest Server

Result:

- Reduces latency drastically ($10 < 50$ ms)
- Improves availability

Conclusion:

The delay is caused by inefficient DNS design. Using Anycast DNS, pre-fetching and optimized TTL can reduce lookup time to below 50 ms.

DNS Case - Study - Impact of centralized DNS Architecture on performance

A centralized DNS architecture where all queries are handled by single primary DNS Server has several serious drawbacks

- High latency: All global request travel to one server $\rightarrow$ Increased delay ($\sim 800ms$)
- Server overload: Heavy traffic causes congestion and slow responses
- Single point of failure: if the server fails all services become unreachable
- Poor scalability: cannot efficiently handle of users

Result: centralized DNS leads to slow response, low reliability and poor scalability.

Distributed DNS Solution Design: To improve performance and availability

Anycast DNS:

- Deploy multiple DNS Servers globally with same IP
- Routes users to the nearest server $\rightarrow$ reduces latency

Benefits:

- Reduces lookup latency significantly
- Automatically reroutes traffic if a node fails

Geo-distributed DNS Servers:

- place servers in different continents
- Reduces distance and improves response time

DNS load Balancing :: Distribute traffic across multiple servers

Redundant (primary + Secondary) DNS :: Ensures service continuity during failures

Integration with CDN :: Use Content Delivery Networks, DNS direct users to nearest content server.

Result :: lookup latency significantly reduced and high availability achieved

Role of TTL, Anycast and caching :

TTL (Time to live)

- **High TTL** → better caching, faster responses
- **Low TTL** → quicker updates but more DNS queries

Anycast Routing :

Directs users to nearest DNS server

Reduces latency and improves fault tolerance

Conclusion :

Together they improve performance, scalability and fault tolerance

CASE STUDY - Stock trading platform

factors causing Latency Increase:

a) Flow control

- Receiver limits sender's data rate
- During high traffic, buffer-bills $\rightarrow$ allows transmission

b) Nagle's Algorithm

- combines small packets before sending
- Introduces delay in real-time transactions

problem in trading system:

- orders are small but time-sensitive
- Nagle delays sending until enough data accumulates
- causes unnecessary waiting time

SYN Queue Limitations:

High connection requests overload SYN queue
causes delays in connection establishment

Effect on High-frequency Transactions

flow control $\rightarrow$ delays order processing due to slower data transfer

Nagle's Algorithm $\rightarrow$ delays sending small order packets

SYN Queue overflow $\rightarrow$ delays or drops connection request

Overall Impact:

- Latency increases from 1ms to 40ms
- Leads to
 - Delayed confirmations
 - Missed trading opportunities
 - customer dissatisfaction

Recommended TCP Improvements:

- Disable Nagle's Algorithm (TCP_NODELAY) → Sends packets immediately → reduces delay
- Increase TCP buffer sizes → Improves flow control handling during peak load
- Optimize SYN queue (Increase backlog) → Handles more connection requests efficiently
- Use persistent connections → Avoid repeated connection setup delays
- Enable TCP Fast Open (Optional) → Reduces connection establishment time

RTT - Reduce Retransmission Timeout

- Detects lost packets faster
- Improves responsiveness

Conclusion:

The latency increase is caused by flow control delays, Nagle's buffering and SYN queue congestion. By optimizing TCP configurations, the system can achieve low latency and high reliability, which is critical for real-time stock trading.